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# Development of compressive strength prediction platform for concrete materials based on machine learning techniques

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ABSTRACT

With the continuous development of artificial intelligence, machine learning (ML), as an important branch, is used to promote the digitalization of concrete. Considering that the verification of the strength corresponding to the concrete mixture requires strict curing system and cycle, high cost and low efficiency, ML technology is employed to predict the 28-day compressive strength of ordinary concrete for the support of engineering practices. The aim is to develop an efficient platform for predicting the compressive strength of concrete materials. Ridge Regression (RR), Least Absolute Shrinkage and Selection Operator (Lasso), Elastic Net Regression (EnR), Support Vector Regression (SVR) and K-Nearest Neighbor (KNN) are established, of which efficiency and accuracy are analyzed and calculated in detail. Then, integrated algorithms, namely Gradient Boosting Decision Tree (GBDT), eXtreme Gradient Boosting (XGBoost) and Random Forest (RF), are used to improve the prediction accuracy of a single machine learning algorithm. Finally, Grid Search (GS) optimization algorithm is used to further improve the generalization performance of the model. The results show that the $R^2$ in training set of the single models is about 0.8, which is obviously lower than that of ensemble models (>0.95). It is encouraging that after conducting grid search optimization, the prediction accuracy of ML models can be improved obviously, of which $R^2$ increasing by around 15%. In addition, the GS-XGBoost model achieved the most excellent prediction accuracy and generalization performance, with $R^2$ values of over 99% in both the training and test sets. Finally, based on the results obtained, a software platform with a convenient graphical user interface (GUI) has been developed, which can greatly improve the efficiency of material design and testing required by researchers and engineers.

## 1. Introduction

Concrete is a commonly used material in modern constructions, such as houses, commercial buildings, roads, and bridges [1–3]. Nowadays, the increasingly complex service environment has made it necessary for concrete materials to pursue higher performance, which motivates the emergence of various new concretes, such as high/ultra-high performance concrete, self-compacting concrete and

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self-healing concrete [4–8]. Meanwhile, the environmental issues of high carbon emissions and high energy consumption also require the concrete sector to move toward green and sustainable development [9–12]. Many researchers recycle various waste materials (e.g., carbonized recycled powder, construction and demolition waste, and phosphogypsum to the concrete material) to design and manufacture green concretes [13–15]. Undoubtedly, these strategies made the concrete system more and more complex. This brings a new challenge in the concrete industry: the performance prediction is becoming increasingly difficult. Strength is one of the most important properties in the concrete system, which is closely associated with the structural design since it is necessary to know the concrete strength before constructing buildings [16,17]. Therefore, it is of great significance to find a suitable strategy for the prediction of concrete performance [18].

There have been many studies focusing on the performance prediction of concrete. In the early days, engineers usually employed empirical equation to predict concrete strength, especially the linear equation [19]. However, as can be understood, these traditional empirical equations are unsuitable for the complex modern concretes [20,21]. Hence, some researchers proposed to use multiple multivariate fitting methods to predict concrete performance, including concrete strength, fluidity, and chloride diffusion coefficient [22–24]. In ordinary concrete systems, the most used performance fitting methods include Response Surface Methodology (RSM), D-optimal mixture design (DOD) and multiple regression model based on mathematical model. RSM is a statistical and experimental design method that helps establish a relationship model between input factors and output responses. Habibi et al. [24]used the RSM method to evaluate the mechanical properties and durability of Recycled Aggregate Concrete (RAC). The results showed that the evaluation parameters ($R^2$) of the fluidity and strength prediction models established by RSM are all above 90%. Ma et al. [25] employed RSM method to predict the compressive strength of manufactured sand concrete considering the multiple factors. SP, PFA and SF were considered as input variables in the model. The significance level of variance analysis of the final results was less than 0.05, and the multivariate correlation coefficient ($R^2$) was 0.85. The DOD method explores the influence of material parameters on strength and establishes a relationship model between material strength and parameters through sampling and regression analysis. Fan et al. [26] established a regression model for predicting the performance of eco-friendly ultra-high performance concrete based on the DOD model, and the results showed that the regression model established based on DOD had a high degree of fit ($R^2>0.9$). Besides, Sergis et al. [27] designed 3D high-performance concrete by the DOD method, and six independent influencing factors including the type of cement, sand and superplasticizer, the ratio of water binder to sand binder, viscosity modifiers and C-S-H seed additives were considered in the study. Compared with RSM model, DOD model shows better performance in terms of model accuracy, complexity of input variables and interaction between input variables. Yu et al. [28] utilized the Bayesian theory and the Markov Chain Monte Carlo (MCMC) method to establish the chloride diffusion coefficient of concrete. This provided an effective method to describe the probability characteristics of chloride diffusion coefficient using probability density function and cumulative distribution function. Homayoonmehr et al. [29] compared the prediction of chloride ion diffusion coefficient based on mathematical regression model with other models, and the surface mathematical statistical regression model has higher accuracy and interpretability. Shafikhani et al. [23] compared the quantification effect of the mathematical statistical model and the mixed model for chloride ion diffusion coefficient, showing that the mixed model had better accuracy, with an accuracy ($R^2$) of 0.97. However, as the complexity of ecological concrete systems increases and the dataset continues to expand, the predictive accuracy and efficiency of these models is increasingly insufficient, limiting their further applications. Therefore, it is necessary to develop new techniques to predict the concrete performance with high accuracy.

With the rising development of Artificial Intelligence (AI) technology, its important branch Machine Learning (ML) has been widely used in various fields, such as minerals industry, aerospace industry, and construction industry [30–32]. Many researchers have also applied machine learning techniques to the prediction of key concrete properties [33–35]. For example, Salami et al. [36] used Artificial Neural Network (ANN), Gene Expression Programming (GEP), and gradient boosting tree (GBT) methods to predict the compressive strength of lightweight foam concrete. The results showed that the GBT model performed better than other models in accuracy, with accuracy rates of more than 0.97 for both training and validation sets. Nguyen et al. [37] selected four models, Support Vector Regression (SVR), Multi-Layer Perceptron (MLP), Gradient Boosting Regression (GBR), and eXtreme Gradient Boosting (XGBoost), to predict the strength of concrete, and the results showed that GBR and XGBoost produced better prediction results with significantly less computational effort than SVR and MLP. Other researchers have compared the effects of traditional multivariate fitting methods and ML algorithms on the prediction of concrete performance, finding that both traditional multivariate fitting methods and ML model were powerful tools for predicting concrete performance. However, compared with traditional fitting methods, ML had higher accuracy. For example, Hammoudi et al. [38] compared the performance of RSM and ANN algorithms in predicting the strength of recycled coarse aggregate concrete, and found that both algorithms were useful, but the ANN prediction accuracy of 0.9980 is closer to 1. Therefore, it is meaningful to use ML techniques for the performance prediction of concrete materials.

To further adapt to the prediction goals of complex systems, researchers proposed optimization algorithms to optimize simple ML algorithm model [39–41]. For example, similarly, Zhao et al. [39] used Bayesian optimization algorithm to optimize the XGBoost model, and used the optimized model to predict the flowability and compressive strength of cement-based grout material, and found that the introduction of Bayesian optimization algorithm improved the accuracy of the XGBoost algorithm and enhanced the generalization ability of the model. Wu et al. [40] utilized grid search algorithm to optimize the SVR machine learning algorithm, and then the established model was employed to predict the compressive strength of sustainable concrete. The Grid Search optimization SVR model (GS-SVR) achieved the better prediction performance with an $R^2$ value of 0.93, surpassing the SVR models with $R^2 = 0.85$. Although the models and optimization models applied to predict the concrete performance are increasing in recent years, for most research, the scope of research is relatively small, and there are few studies on the horizontal expansion of various types of concrete and the vertical comparison of algorithmic levels. Therefore, this paper summarizes and evaluates the commonly used algorithm models to provide a deeper understanding for predicting concrete performance by using ML techniques.

This study aims to establish an efficient and accurate model for concrete performance prediction, and develop a platform for assessing key performance indicators based on this model. At first, single ML predictive models, i.e., Ridge Regression (RR), Least Absolute Shrinkage and Selection Operator (Lasso), Elastic Net Regression (EnR), Support Vector Regression (SVR) and K-Nearest Neighbor (KNN) are established, of which efficiency and accuracy are analyzed and calculated in detail. Then, ensemble algorithms, viz., Gradient Boosting Decision Tree (GBDT), eXtreme Gradient Boosting (XGBoost), and Random Forest (RF), were employed to enhance the prediction accuracy of a single machine learning algorithm. Finally, the grid search optimization algorithm is employed to further enhance the model's generalization performance. In addition, the different methods for predicting the strength of ordinary concrete are discussed, including overcoming prediction errors, and improving prediction accuracy. To conclude, this study presents novel ideas and methodologies for investigating the strength prediction of ordinary concrete, which can be used as a valuable reference in related fields.

## 2. Database analysis

The dataset used in this study is obtained from the CCCC Second Harbor Engineering Company LTD, which contains a total of 487 data. The dataset considers eight input variables: cement (C), fly ash (F), mineral powder (MP), admixture, fine aggregate (FA), coarse aggregate (CA), water (W), and superplasticizer (SP), with strength as the output variable. The characteristics of cement, mineral powder and fly ash are shown in Table 1, and the basic characteristics of aggregate are shown in Table 2. Ordinary Portland cement P.II 42.5 was used as the test cement. The 28d activity index of mineral powder and fly ash ranged from 95 to 122% and 71–97%, respectively. Fine aggregate includes machine-made sand and natural sand. As can be seen in Table 1, the fineness modulus of machine-made sand and natural sand are 2.4-3.3 and 2.5-3.1 respectively, and the mud content of natural sand is 1.2-2.6. It can be seen from the information given in Table 1 that the coarse aggregate contains three particle sizes of gravel, which are 5-10 mm, 10–20 mm and 16-31.5 mm respectively, and the corresponding void ratio ranges are 37-45%, 38–45% and 42–44% respectively. The superplasticizer is a high performance polycarboxylic acid superplasticizer with an average solid content of 20.8%.

During the curing process of the sample, standard curing procedures were employed. The curing temperature was maintained at 20 $\pm$ 2 °C, with a relative humidity of no less than 95%. The strength is tested in accordance with the standard (GB/T 50081-2019) [42], and the test block used in the test is 150 $\times$ 150 $\times$ 150 mm. According to the standard (GB/T 50080-2016) [43], the workability of fresh concrete was tested. According to the requirements of relevant specifications and actual projects, the general range of collapse is determined to be 40–220 mm.

In this dataset, the parameter range, standard deviation, and average values are shown in Table 3. The distribution histograms of input and output variables are shown in Fig. 1. Most of the input variables exhibit a normal distribution, which meets the requirements of ML for input variables [44], thus ensuring the effectiveness of machine learning algorithms based on this data. The impact on strength is limited when the admixture yield is small. Additionally, the correlation analysis of the database in **Section 4.1** reveals a low correlation coefficient between mixture and strength. Therefore, utilizing this dataset where the admixture is not normally distributed has minimal impact on overall strength prediction.

## 3. Machine learning techniques

### 3.1. Data preprocessing

Based on outlier detection techniques, the database is subjected to certain cleansing procedures [45]. Due to significant differences in the magnitudes of input variables, asymmetric impacts on the predictive results may occur. Therefore, the database is subjected to min-max normalization on each column to achieve normalization [46]. The whole data are normalized using **Eq. (1)**:

$$x' = \frac{x - x_{min}}{x_{max} - x_{min}} \tag{1}$$

where $x$ represents a raw data value, while $x_{min}$ and $x_{max}$ represent the minimum and maximum values, respectively, within the column where $x$ is located.

### 3.2. Assessment criteria

Four regression indicators, including R-squared ($R^2$), Mean Absolute Error (MAE), Root Mean Square Error (RMSE), Mean Square Error (MSE) and Mean Absolute Percentage Error (MAPE) are considered to evaluate the accuracy of algorithms [47,48].

#### 3.2.1. $R^2$

$R^2$ represents the goodness of fit of the regression model and ranges from 0 to 1. A value close to 1 indicates a very accurate fit of the

**Table 1**
Characteristics of Cementitious material.

| Cement | | Mineral powder | | Fly ash | | |
|---|---|---|---|---|---|---|
| Density (g/cm$^3$) | Specific surface area (m$^2$/g) | Activity index (%) | Specific surface area (m$^2$/g) | Activity index (%) | Density (g/cm$^3$) | 45 μm sieve residue |
| 2.98-3.16 | 304-414 | 95-122 | 401-576 | 71-97 | 2.25-2.58 | 2.5-29.2 |

**Table 2**
Characteristics of aggregate.

| Void ratio of coarse aggregate (%) | | | Fineness modulus of fine aggregate | |
|---|---|---|---|---|
| Gravel 5–10 mm | Gravel 10–20 mm | Gravel 16-31.5 mm | Machine-made sand | Natural sand |
| 37-45 | 38-45 | 42-44 | 2.4-3.3 | 2.5-3.1 |

**Table 3**
Description of the input and output data of the predicted intensity.

| Variable | Symbol | Category | Statistics | | | |
|---|---|---|---|---|---|---|
| | | | Min | Max | Average | STDEV |
| Cement | C (kg/m$^3$) | input | 174.0 | 494.0 | 309.0 | 70.7 |
| Fly Ash | Fly ash (kg/m$^3$) | input | 0 | 137.0 | 64.0 | 37.4 |
| mineral powder | MP (kg/m$^3$) | input | 0 | 182.0 | 14.3 | 31.2 |
| admixture | Admixture (kg/m$^3$) | input | 0 | 34.0 | 0.1 | 2.1 |
| Fine aggregate | FA (kg/m$^3$) | input | 635.0 | 935.0 | 773.0 | 54.1 |
| Coarse aggregate | CA (kg/m$^3$) | input | 996.0 | 1241.0 | 1082.6 | 40.5 |
| Water | Water (kg/m$^3$) | input | 102.0 | 185.0 | 156.3 | 7.5 |
| Superplasticizer | SP (kg/m$^3$) | input | 0 | 12.8 | 4.2 | 1.6 |
| strength | STRENGTH (MPa) | output | 21.6 | 73.3 | 45.8 | 11.5 |

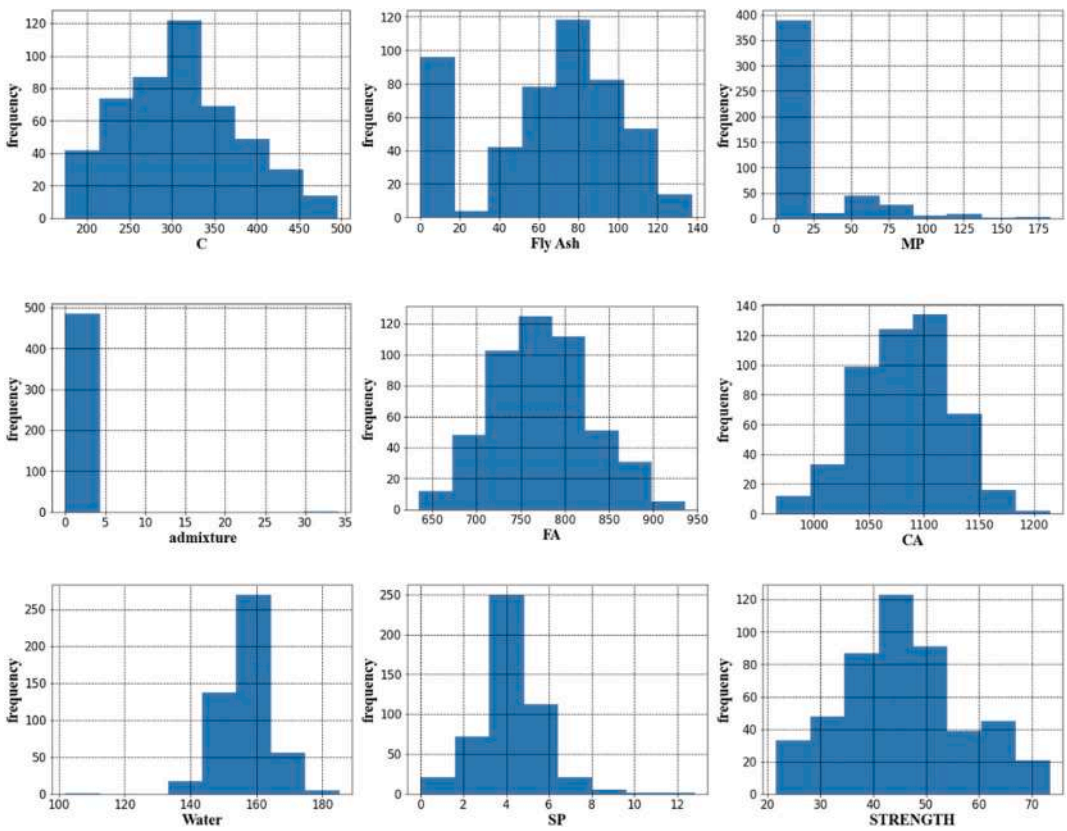


**Fig. 1.** Histogram plot displaying the distribution of the dataset.

model, while a value close to 0 indicates a poor fit, which can be calculated by Eq. (2):

$$R2 = 1 - \sum_{i=1}^{n} \left(y_i - y_{prei}\right)^2 \Big/ \sum_{i=1}^{n} \left(y_i - \bar{y}\right)^2 \tag{2}$$

where $y_i$ is the numerical value of the i-th observed intensity, $y_{prei}$ is the value of the i-th estimated using model, $\bar{y}$ is the mean value of

calculated strength data.

### 3.2.2. MAE

MAE represents the arithmetic mean of the deviations, which can be figured out by Eq. (3):

$$MAE = (1/n)\sum_{i=1}^{n}\left|y_i - y_{prei}\right| \tag{3}$$

### 3.2.3. MSE

MSE stands for mean squared error, which is the average of the sum of squares of errors between predicted and original data points defined as Eq. (4):

$$MSE = (1/n)\sum_{i=1}^{n} w_i\left(y_i - \hat{y}_i\right)^2 \tag{4}$$

### 3.2.4. RMSE

RMS or Root Mean Square Error, is a commonly used loss function in regression analysis. It is calculated as the square root of the mean squared error, as shown in Eq. (5):

$$RMSE = \sqrt{(1/n)\sum_{i=1}^{n}\left(y_i - y_{prei}\right)^2} \tag{5}$$

### 3.2.5. MAPE

MAPE is the average absolute percentage error, which is the average proportion of the absolute difference between the predicted value and the true value to the true value. It is used to measure the accuracy of the regression model. The smaller the value, the better. which can be calculated by Eq. (6):

$$MAPE = \frac{100\%}{n}\sum\nolimits_{i=1}^{n}\left|\left(y_{prei} - y_i\right)/y_i\right| \tag{6}$$

## 3.3. ML models

The strength of regular concrete can be predicted through a combination of simple and ensemble ML algorithms. Specifically, five basic algorithms- RR [49], LASSO [44], EnR [50], SVR [51], and KNN [52] - form the foundation of the prediction model. To improve model accuracy, four ensemble learning algorithms were incorporated - GBDT [53,54], XGBoost [55] and RF [56], along with two optimization learning algorithms-grid search [57,58]. This combination of algorithms enables a more sophisticated and nuanced approach to predict the compressive strength of ordinary concrete, by considering a wider range of factors and variables that affect the strength and stability of concrete materials.

### 3.3.1. Single methods

Ridge Regression (RR) is a regularization algorithm used to handle linear regression problems [59]. The algorithm controls the complexity of the model by introducing an $L_2$ regularization term to prevent overfitting. It is a complement to the least squares method. The loss function is expressed as follows:

$$\mathrm{J} = (1/n)\sum_{i=1}^{n}\left(f(x_i) - y_i\right)^2 + \lambda \parallel w\parallel_2^2 \tag{7}$$

where $\lambda \parallel \boldsymbol{w}\parallel_2^2$ is the $L_2$ regularization factor.

The LASSO model is an algorithm proposed by Tibshirani in 1996 [60]. The fundamental concept is to minimize the sum of squared residuals while subject to a constraint that limits the sum of absolute values of regression coefficients to a predetermined constant. This approach permits the generation of some regression coefficients that are strictly equal to zero, thereby enabling the creation of a more robust model with stronger explanatory capabilities. This approach bears resemblance to Ridge regression, in which regularization coefficients are introduced in comparison to linear regression. However, LASSO regression differs from Ridge regression in that it employs $L_1$ regularization [61]. The loss function is expressed by Eq. (8):

$$\mathrm{J} = \frac{1}{n}\sum_{i=1}^{n}\left(f(x_i) - y_i\right)^2 + \lambda\|w\|_1 \tag{8}$$

where $\lambda\|\boldsymbol{w}\|_1$ is the $L_1$ regularization factor.

The Elastic Net Regression Algorithm (EMRA) is a combination of the ridge regression algorithm and the LASSO algorithm. This algorithm combines the advantages of the regularization of both regression algorithms with a loss function controlled by two penalty coefficients. The loss function is expressed by Eq. (9):

$$J(w) = \sum_{i=1}^{n} \left(y_i - w^T x_i\right)^2 + \lambda\rho\|w\|_1 + \|w\|_2^2(\lambda(1-\rho)) \Big/ 2 \tag{9}$$

where $\rho$ is the two penalty coefficients of the elastic net algorithm. When $\rho = 0$, its cost function is equivalent to that of Ridge regression, and when $\rho = 1$, its cost function is equivalent to that of Lasso regression. The ENRA retains the nature of feature selection of the LASSO algorithm, while considering the stability of ridge regression.

SVR is an important branch of SVM and is an important algorithm used in SVM for regression, the purpose of SVR is to find a hyperplane so that the total deviation of all sample points from the hyperplane is minimized [51]. The main principle is to create an "interval band" on both sides of the linear function, the distance from the hyperplane is "ε". All samples falling in the interval band are not counted as loss, and finally the optimized model is obtained by minimizing the total loss and maximizing the interval.

The core thinking of KNN algorithm is that samples with higher similarity, mapped to the n-dimensional space, are closer in distance to avoid samples with lower similarity. KNN algorithm can be used for both classification and regression problems, and when it is used for regression calculations, the mean (or weighted mean) of K neighbors is generally used, as the prediction result.

### 3.3.2. Ensemble methods

*3.3.2.1. GBDT algorithm.* The Gradient Boosted Decision Tree (GBDT) is a machine learning algorithm that combines the Gradient Boosting Regression algorithm and the decision tree algorithm. During each iteration, GBDT learns the decision tree by fitting negative gradients (i.e., residuals). Specifically, GBDT calculates residuals based on the predicted results of the current model and the true labels, and then leverages the next weak classifier to fit these residuals, thereby continuously enhancing the performance of the model [62]. The advantages of the algorithm are its ability to handle high-dimensional sparse data, its ability to automatically perform feature selection and handle problems such as missing values, and its good performance in dealing with nonlinear, nonconvex, and nondistributed large-scale data.

Define the input data set as $T = \{(x_1,y_1),(x_2,y_2),\cdots,(x_n,y_n)\}, x_i \in X \subseteq R^n, y_i \in Y \subseteq R$, the loss function is $L(y,f(x))$, and the output is the regression tree $\widehat{f}(x)$. The GBDT algorithm works in the following three steps [63]. The workflow of the GBDT model is shown in Fig. 2 and the principle is as following:

Step 1: Initialize the weak learner.

$$f_0(x) = \underset{c}{\arg min} \sum_{i=1}^{N} L(y_i, c) \tag{10}$$

Step 2: Calculate the negative gradient, also called the residual.

$$r_{mi} = -\left[\partial L(y_i, f(x_i))/\partial f(x_i)\right]_{fx)=f_{m-1}(x}\Big) \tag{11}$$

Step 3: get the final GBDT model.

$$\widehat{f}(\boldsymbol{x}) = f_M(x) = f_0(x) + \sum_{m=1}^{M}\sum_{j=1}^{J} c_{mj} I\left(x \in R_{mj}\right) \tag{12}$$

*3.3.2.2. XGBoost algorithm.* XGBoost is an ensemble algorithm based on GBDT, proposed by Chen et al., in 2016 [64]. This is a ML algorithm based on gradient boosting decision tree (GBDT) [62]. Comparecd with the GBDT algorithm, the second-order derivative used in the XGBoost algorithm can make the loss function more accurate, while the regularization term can avoid overfitting. Its core idea is to use multiple decision trees to build an ensemble model and combine them to improve prediction accuracy. In the XGBoost algorithm, each decision tree is obtained through iterative training. In each iteration, the algorithm calculates the loss function of the

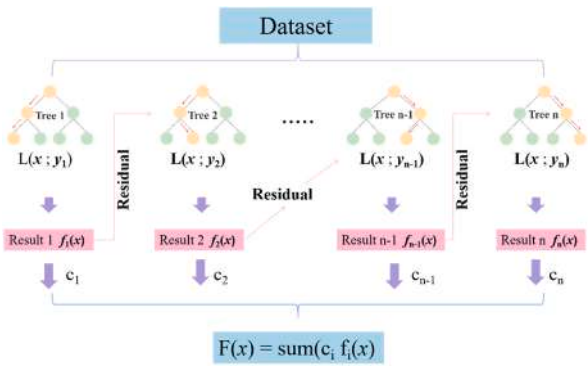


**Fig. 2.** Schematic view of GBDT algorithm.

current model and trains a new decision tree. Afterwards, the new decision tree is designed to correct the errors of the previous decision trees to minimize the loss function of the entire ensemble model [65]. The workflow of the GBDT model is shown in Fig. 3, The objective function of the XGBoost model is expressed by **Eq. (13)**:

$$OBj^{(t)} = \sum_{i=1}^{n} l\left(y_i, \bar{y}_i^{(t-1)} + f_t(x_i)\right) + \Omega(f_t) + \sum_{i=1}^{t-1} \Omega(f_i) \tag{13}$$

*3.3.2.3. RF algorithm.* Random Forest (RF) algorithm is a typical representative of bagging ensemble algorithm. Like GBDT algorithm, RF consists of multiple trees, and the result is determined by the ensemble algorithm model of multiple trees together. Unlike GBDT, RF model can be parallelly generated during the training process, and the result is determined by the vote of multiple trees, which can reduce the variance of the model. Bagging algorithm, also known as bootstrap aggregation algorithm, is a typical algorithm in ensemble learning [56,66]. It consists of two parts, Bootstrap and Aggregating, and was first proposed by Breiman [67] in 2001. Bagging algorithm can be combined with other classification and regression algorithms to improve their accuracy and stability. At the same time, it could avoid overfitting by reducing the variance of the results. The workflow of the RF model is shown in Fig. 4 and the general process of RF algorithm is as follows [68]:

(1) For a given training sample S, M training samples are randomly sampled with replacement each round for n rounds, resulting in n sample sets. It is important to note that these n training sets are independent of each other.
(2) After obtaining the sample sets, a prediction model is obtained each time using one sample set. For n sample sets, we can obtain n prediction models in total.
(3) If the problem to be solved is a classification problem, the classification results can be obtained by voting on the n models obtained earlier; for regression problems, the method of calculating the model average can be used as the final prediction result.

### 3.4. Cross-validation

K-Fold Cross-Validation is a widely used technique for assessing the performance and generalization ability of a given model. Homayoonmehr has shown in the study that 5-fold or 10-fold cross-validation can significantly improve the accuracy and reliability of the model without adding too much computation time. Similar results have been shown in many other studies as well [29,69–71]. In this study, the five-fold cross-validation method is employed, which involves randomly partitioning the original dataset into five nonoverlapping subsets. Subsequently, one of the subsets is used as the validation set, while the remaining four subsets are combined to form the training set, which is then used to train the model. This process is repeated five times, with each subset serving as the validation set exactly once, and the results are averaged to provide an overall estimate of the model's performance [72]. This process is repeated 5 times, each time using a different subset as the validation set and the remaining 4 subsets as the training set [51]. The average result of the five experiments is used as the performance metric of the model. five-Flod Cross-validation is shown in Fig. 5.

The advantage of K-fold cross-validation is that the performance of the model can be evaluated more accurately, especially when the dataset is small. It can use the data efficiently because all the data will be used to train and test the model. In addition, K-fold cross-validation can also help avoid overfitting problems.

### 3.5. Optimization algorithm

The optimization algorithm used in this paper is the randomized grid search algorithm. GridSearchCV algorithm is a parameter tuning algorithm based on grid search [73]. It can automatically search through a given parameter space to find the best combination of parameters, improving model performance and generalization ability [74]. In this algorithm, a parameter grid is firstly defined, which is the range of values for given parameters. Then, the cross-validation is used to evaluate the performance of each parameter

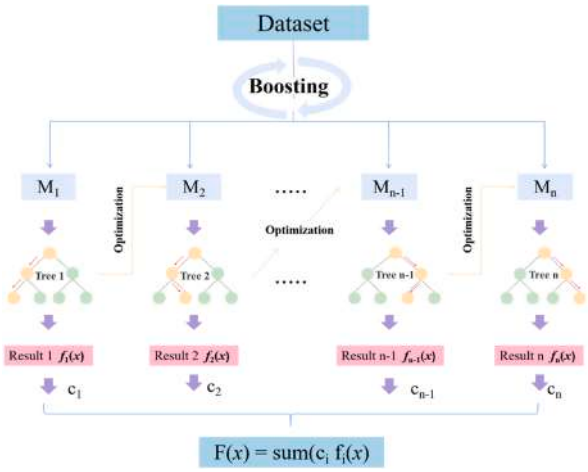


**Fig. 3.** Schematic view of XGBoost algorithm.

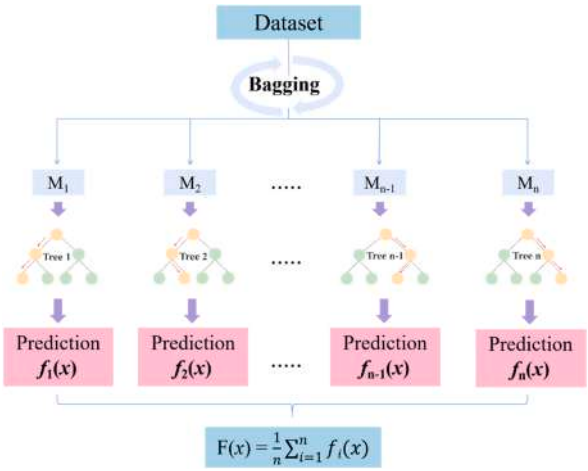


**Fig. 4.** Schematic view of RF algorithm.

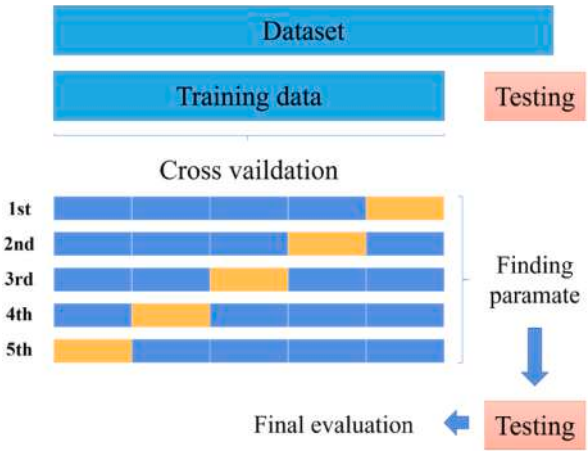


**Fig. 5.** Schematic description of five-fold validation.

combination to select the one with the best performance. This process can be applied to various machine learning algorithms such as support vector machines, decision trees, random forests, etc [54,58,75].

The workflow of GridSearchCV for hyperparameter optimization is as follows:

(1) Define parameter grid: Create a parameter grid that contains all the parameters and their value ranges to be searched by defining the parameter ranges.
(2) Create a model: Create a model based on the type of algorithm to be optimized and pass it as a parameter to GridSearchCV.
(3) Cross-validation: Use cross-validation technique to split the dataset into multiple parts and alternate each part as the test set and training set. For each model and parameter combination, evaluate the model's performance using cross-validation and calculate the mean accuracy score.
(4) Find the best parameter combination: GridSearchCV will automatically search all possible parameter combinations and return the best parameter combination with the highest mean accuracy score, thus improving the model performance and generalization ability.
(5) Apply the best parameter combination: Apply the best parameter combination to the model, retrain the model, and evaluate its performance using the test set.

## 4. Results and discussions

### *4.1. Feature parameter correlation analysis*

Fig. 6 shows the Pearson correlation coefficients between the input and output variables, which can indicate the positive and negative correlations between the different variables. The study focuses on the correlation between compressive strength and mix proportion in concrete, and therefore uses the Pearson correlation coefficients algorithm for univariate correlation analysis [76]. From

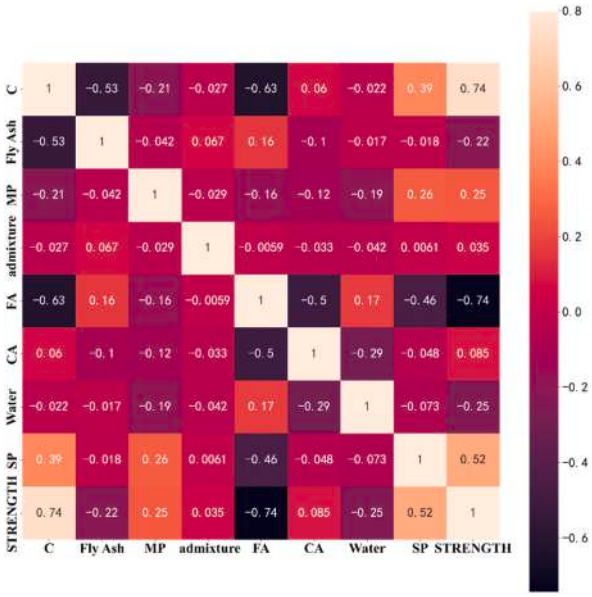


**Fig. 6.** Pearson correlation coefficient matrix.

Fig. 6, it can be seen that the correlation coefficients between all input variables and strength are less than 0.95, indicating that there is a non-linear relationship between strength and input variables [77]. Among the eight input variables, a significant positive correlation was observed between cement content and strength output (correlation coefficient = 0.74), while a significant negative correlation was found between fine aggregate content and strength output (correlation coefficient = -0.74). Moreover, the compressive strength was positively correlated with mineral powder, water content, and fly ash at coefficients of 0.25, 0.25, and 0.22, respectively. The positive correlation between cement and strength is primarily attributed to the fact that the hydration product of cement, C-S-H gel, not only serves as the main crystalline component in concrete for providing strength but also functions as a highly dense C-S-H gel capable of filling micropores and capillaries within concrete, thereby enhancing its compactness [78]. The strength of concrete exhibits a strong negative correlation with the amount of fine aggregate. This could be attributed to the effects of ingredient grad on the packing structure of concrete. Specifically, when q falls within the range of 0.21-0.25, concrete displays superior mechanical properties. Excessive amounts of fine or coarse aggregate can result in a less dense matrix skeleton, ultimately leading to decreased compressive strength [79]. For this dataset, the addition of fine aggregate is significantly higher than the optimal grade, thus showing a negative correlation, so it is recommended to reduce the amount of fine aggregate to achieve the enhanced effect. The positive correlation between mineral powder and fly ash and strength is primarily due to the presence of reactive substances within them that can undergo hydration reactions, resulting in the production of cementitious compounds. Additionally, mineral powder and fly ash contain a significant proportion of fine-grained particles that can fill voids between aggregates. The impact of different materials on strength is highly intricate, necessitating the use of artificial intelligence techniques for predictive modeling.

### 4.2. Comparison of the models

#### 4.2.1. Single methods

During model training, the dataset is randomly split into a training set and a test set, with the training set accounting for 80% of the total dataset and the test set accounting for 20%. Table 4 presents the $R^2$, MSE, RMSE, MAE, and MAPE values for five single models (i.e., RR, LASSO, EnR, SVR and KNN) to facilitate comparative analysis of their predictive performance. Fig. 7 shows the distribution of predicted values and true values for five single models. From Table 4, the KNN model has the highest fit coefficient with $R^2 = 0.8748$ compared with the other models in the training set. The values of the other four evaluation metrics MSE, RMSE, MAE, and MAPE are 15.9150, 3.9893, 3.0411, and 7.1656, respectively, which are the smallest values among the single models. The SVR model exhibited

**Table 4**
Statistical modeling results for single model numerical systems and related systems.

| Model | Training data | | | | | Testing data | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | $R^2$ | MSE | RMSE | MAE | MAPE | $R^2$ | MSE | RMSE | MAE | MAPE |
| RR | 0.7829 | 28.7644 | 5.3632 | 4.1716 | 9.7385 | 0.7908 | 31.3944 | 5.603 | 4.3760 | 9.7385 |
| Lasso | 0.7941 | 27.2796 | 5.2229 | 4.068 | 9.5099 | 0.8084 | 28.8146 | 5.603 | 4.1990 | 9.5009 |
| EnR | 0.7941 | 27.2847 | 5.2234 | 4.0684 | 9.5099 | 0.8080 | 28.8146 | 5.603 | 4.2018 | 9.5009 |
| SVR | 0.5678 | 57.274 | 7.5679 | 5.7534 | 13.0353 | 0.5865 | 62.0565 | 5.603 | 4.1172 | 13.053 |
| KNN | 0.8748 | 15.9150 | 3.9893 | 3.0411 | 7.1656 | 0.7921 | 31.2004 | 5.5857 | 4.6936 | 7.1656 |

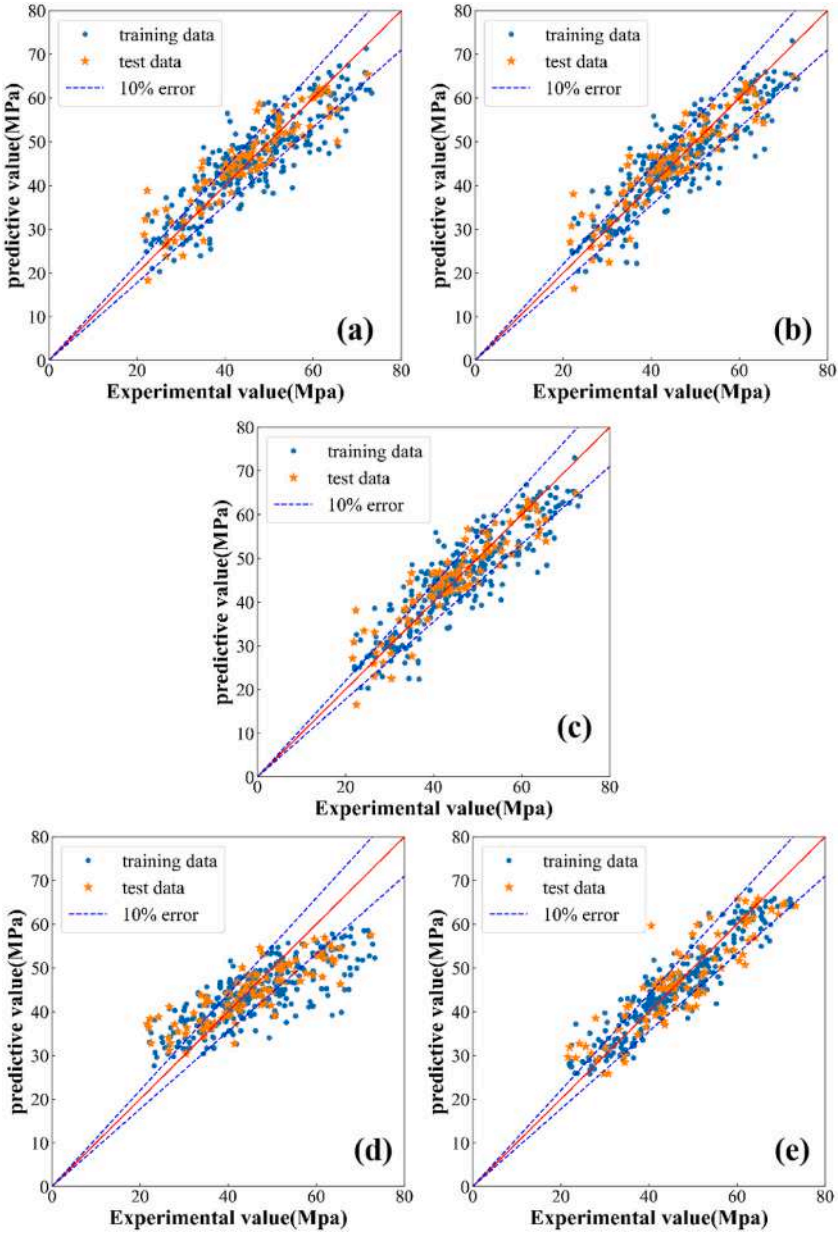


**Fig. 7.** Relationship between actual and predicted values of compressive strength for five individual models: (a) RR, (b) LASSO, (c) EnR, (d) SVR, (e) KNN.

the lowest coefficient of fit, with $R^2 = 0.5678$, as opposed to the other models. Furthermore, the other four evaluation metrics, namely MSE, RMSE, MAE, and MAPE, had values of 57.274, 7.5679, 5.7534, and 13.0353, respectively, which were the highest values obtained by any of the individual models. The evaluation indicators of the remaining models fall within the range defined by the two models. The distribution depicted in Fig. 7 indicates that the KNN model data points are predominantly clustered within and proximate to the 10% error line, while the SVR model exhibits a significant deviation from the 45° line. The synthesis indicates that among the five single models employed for predicting concrete strength, the KNN model exhibits superior applicability while the SVR model is less effective. This can be attributed to the fact that in regression algorithms, the KNN model calculates an average of k points near a given value as its final output, which proves more advantageous when predicting strength with varying raw materials [80]. However, it is undeniable that the fitting of the KNN model on the test set is $R^2 = 0.7921$, and its other four evaluation metrics (MSE, RMSE, MAE, MAPE) are equivalent to those of the RR, EnR, and LASSO models, which suggests a weak generalization ability for the KNN model.

Additionally, a quantitative analysis was conducted on the prediction residuals. Fig. 8 displays frequency distribution histograms and kernel density plots of the prediction residuals for each of the five single models. From the figure, it is evident that the mean values of RR, EnR, LASSO and KNN models for the training set are centered around 0. The comparison of variance σ reveals that KNN model

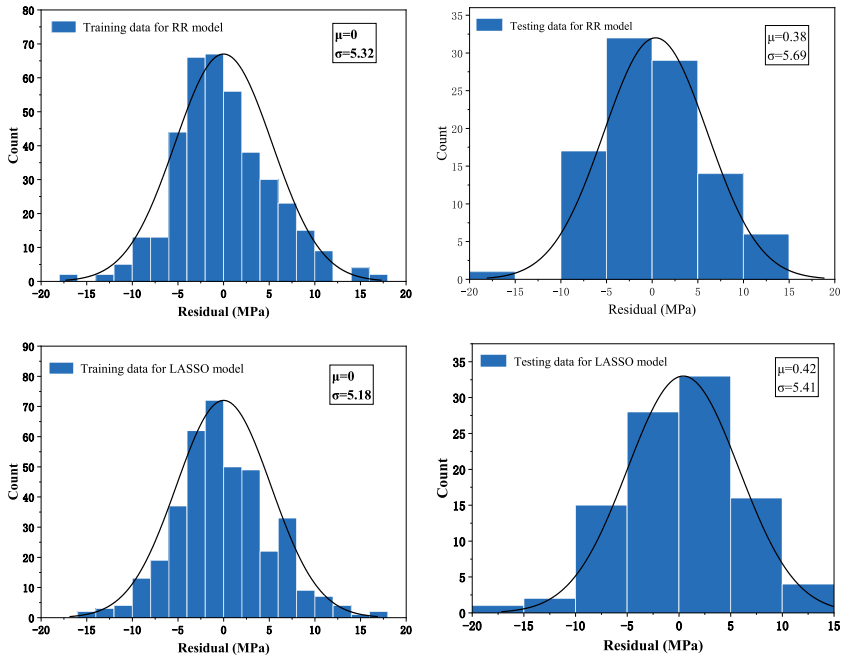


**Fig. 8.** Residual of five individual models.

has the lowest value of σ = 3.99 while SVR model exhibits a mean value μ = 1.49 and variance σ = 7.85, which deviates from standard normal distribution. However, the residual distribution results for the test set clearly indicate that the well-fitted KNN model has poor performance, suggesting limited cross-generalization ability. The mean μ and variance σ of the more frequently used KNN model are 0.23 and 5.58, respectively, which is similar to other models. Both the training set and test set yield results consistent with those of the evaluation metrics in Table 4, further confirming that models utilizing hyperplanes for regression (e.g., SVR) are less suitable for performance prediction of ordinary concrete. Conversely, models based on proximity principles (e.g., KNN) are more applicable but less generalizable. Therefore, improving the generalization performance of simple models is an important task.

### *4.2.2. Ensemble methods*

The performance metrics, including $R^2$, MSE, MASE, MAE and MAPE of the ensemble models (GBDT, XGBoost and RF), are presented in Table 5. Additionally, Fig. 9 illustrates the distribution between predicted and true values for the combined models. As depicted in Fig. 9, the training data points of both XGBoost and RF models are clustered around the 45° line and within the 10% error margin. In contrast, the data points of GBDT model exhibit more dispersion compared to the other two models and partially extend beyond the 10% error boundary. Nevertheless, overall distribution patterns of all three models appear to be more linear than that of a single model shown in Fig. 7. The integrated model demonstrates superior predictive performance compared to the single model, with XGBoost exhibiting the highest accuracy. This is further supported by Table 5, which shows that the fit coefficients of all three models are above 0.95, and XGBoost shows superior performance with values of $R^2 = 0.9984$, MSE = 0.2045, RMSE = 0.4521, MAE = 0.1381, and MAPE = 0.3018, respectively. However, it is undeniable that the distribution of test data points depicted in Fig. 9 lies predominantly outside the 10% error threshold, with some points deviating significantly from said threshold. Additionally, Table 5 reveals that the fit coefficients $R^2$ for the test set fall below 0.85. This suggests that even the XGBoost model, which exhibits the highest level of fit among the ensemble models, experiences overfitting in a similar manner to the KNN model among single models.

Moreover, the prediction residuals of the ensemble models were quantitatively analyzed, as shown in Fig. 10. Fig. 10 shows the frequency distribution histograms and kernel density plots of the predicted residuals for the three integrated models. It is evident from Fig. 10 that the mean value of prediction residuals for the training set of the three ensemble models is approximately 0, and the variance σ is less than 2.5, which is significantly lower compared to the prediction residuals of a single model in Fig. 8. The XGBoost model has the smallest variance of prediction residuals at σ = 1.79, which aligns with results from evaluation index in Table 5. The variance σ of the prediction residuals for all three model prediction sets exceeds 4.5, and the mean value μ is no longer centered around 0, which is consistent with the distribution of test data points in Fig. 9. This suggests that there are discrepancies between predicted and true values for the test set across all three models. The residuals data suggest that the training set of the integrated model is overfitted, while the test set is poorly fitted. This further indicates that there exists an overfitting phenomenon in the ensemble model.

### *4.2.3. Comparison of all models*

It is evident that the overall evaluation performance of the ensemble models is superior to that of the single models. However, it is undeniable that the KNN model has a significantly better predictive effect than other models. This is mainly because in the ordinary

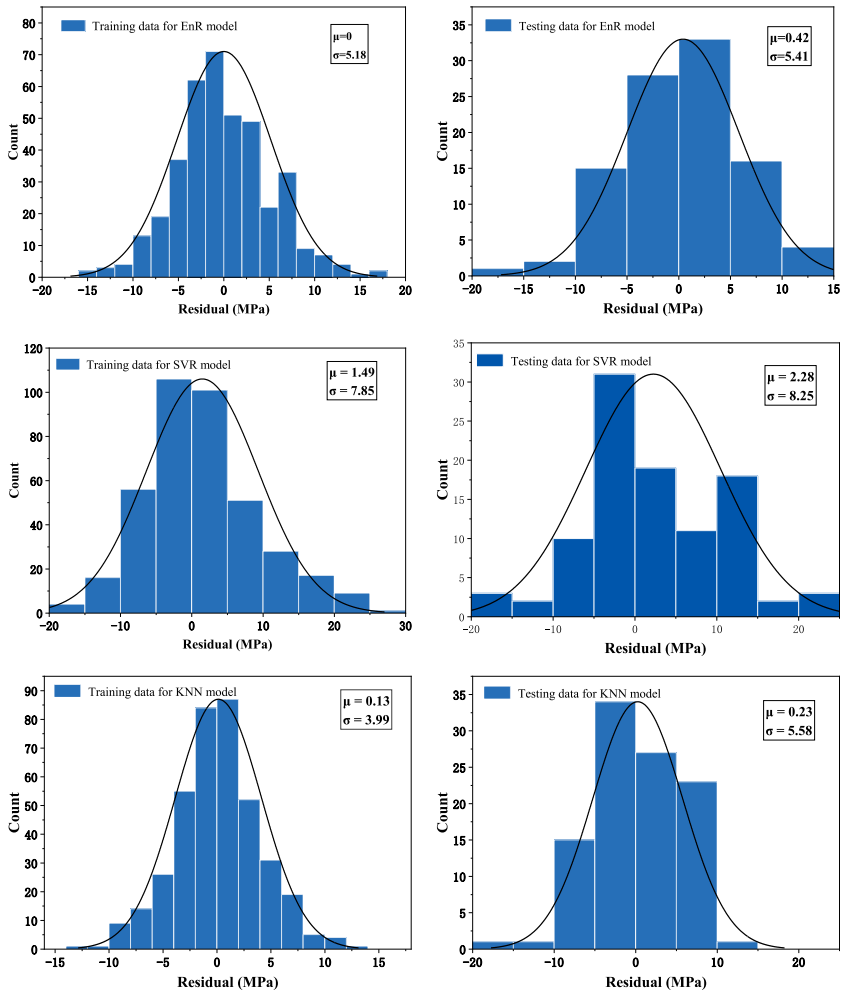


**Fig. 8.** (*continued*).

**Table 5**
Statistical modeling results for ensemble model numerical systems and related systems.

| Model | Training data | | | | | Testing data | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | $R^2$ | MSE | RMSE | MAE | MAPE | $R^2$ | MSE | RMSE | MAE | MAPE |
| GBDT | 0.9523 | 6.0595 | 2.4616 | 1.8449 | 4.3933 | 0.8471 | 22.9468 | 4.7902 | 3.8006 | 4.3933 |
| XGBoost | 0.9984 | 0.2045 | 0.4521 | 0.1381 | 0.3018 | 0.8475 | 22.8830 | 4.7836 | 3.2213 | 3.5351 |
| RF | 0.9785 | 2.7248 | 1.6507 | 1.1760 | 2.8133 | 0.8498 | 22.5321 | 4.7468 | 3.7106 | 2.8133 |

concrete system, strength is a continuous variable that is influenced by multiple factors [81]. High-order KNN models combine the values of the K nearest points based on the proximity principle to predict the strength value. This is consistent with the continuity distribution of strength, so in the single model, the KNN model is more suitable for predicting the performance of ordinary concrete. Fig. 11 shows the radar chart of the evaluation indicators of the eight models. From Fig. 11(a), it is evident that the values of MSE, RMSE, MAE, and MAPE of the training set of the ensemble models are significantly smaller than those of the single models, and the value of $R^2$ is significantly higher than that of the single models. Hence, with an increase in model complexity, the accuracy also increases. This is mainly because the ensemble model combines the prediction results of multiple underlying models and takes advantage of the strengths of different underlying models to compensate for the deficiencies of single models [82]. Compared to a single model, the ensemble approach can better handle data noise and outliers, enhancing the overall stability and robustness [83]. From Fig. 11(b), the evaluation indicators of the testing set of the ensemble models are better than those of the single models, but the advantage is not obvious compared to the training set. This indicates that the generalization performance of the model has been improved to some extent with the increase in model complexity, but the improvement effect is limited [84]. Therefore, the KNN, GBDT, XGBoost, RF models are most suitable for ordinary concrete strength prediction.

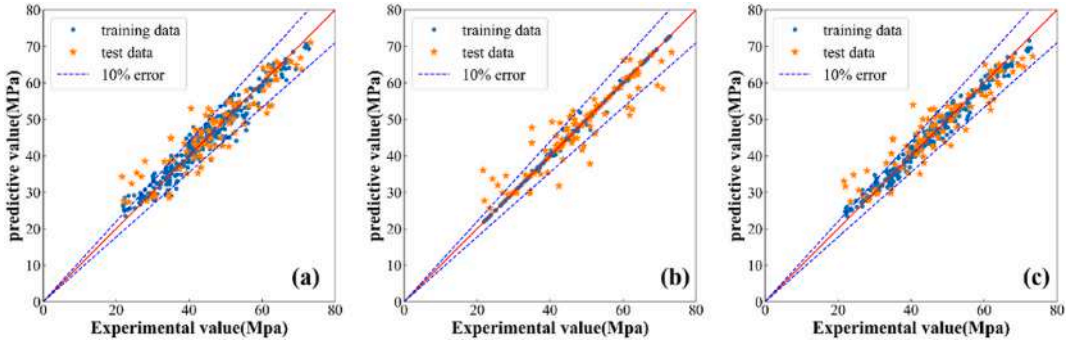


**Fig. 9.** Relationship between actual and predicted values of compressive strength for three ensemble models. (a) GBDT, (b) XGBoost, (c) RF.

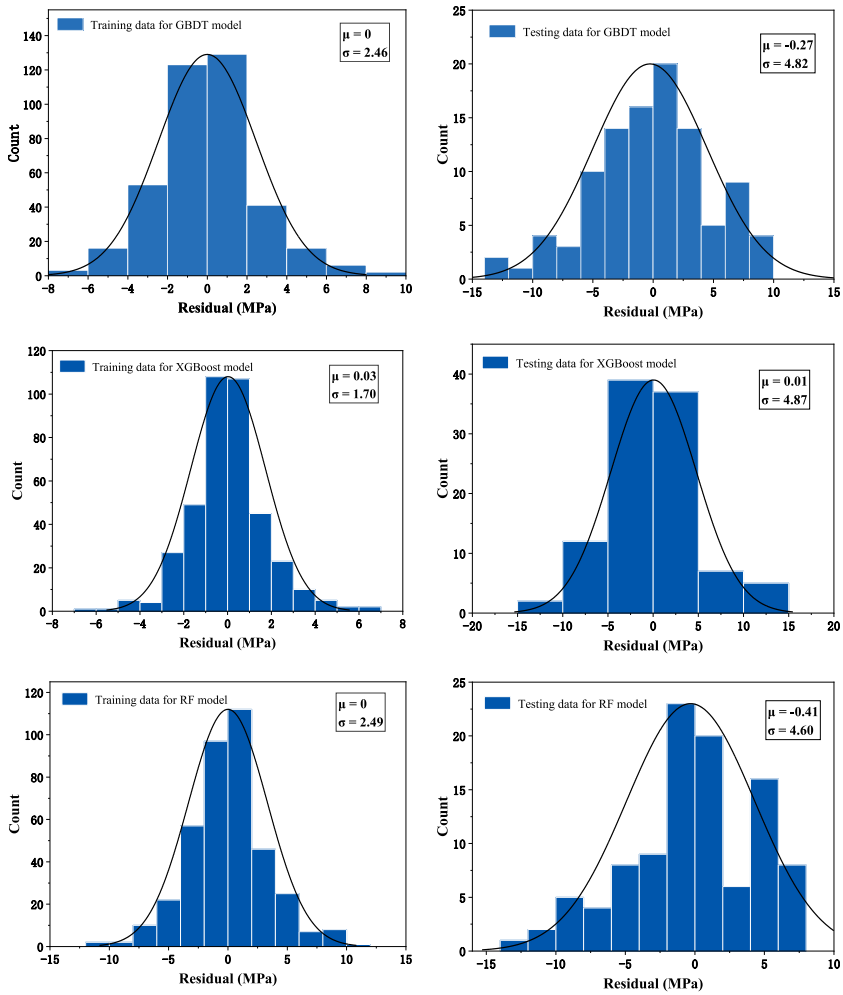


**Fig. 10.** Residual of three ensemble models.

### 4.3. Hyperparameter optimization

To further improve the generalization ability of the models, a stochastic grid search optimization algorithm is utilized to optimize the KNN model and three ensemble models that perform well in a single model. The optimized models are represented by GS-KNN, GS-GBDT, GS-XGBoost, and GS-RF, respectively.

#### 4.3.1. Hyperparameter selection

Table 6 lists the hyperparameters and their scanning ranges for four models. The hyperparameter selected by KNN is n-neighbors,

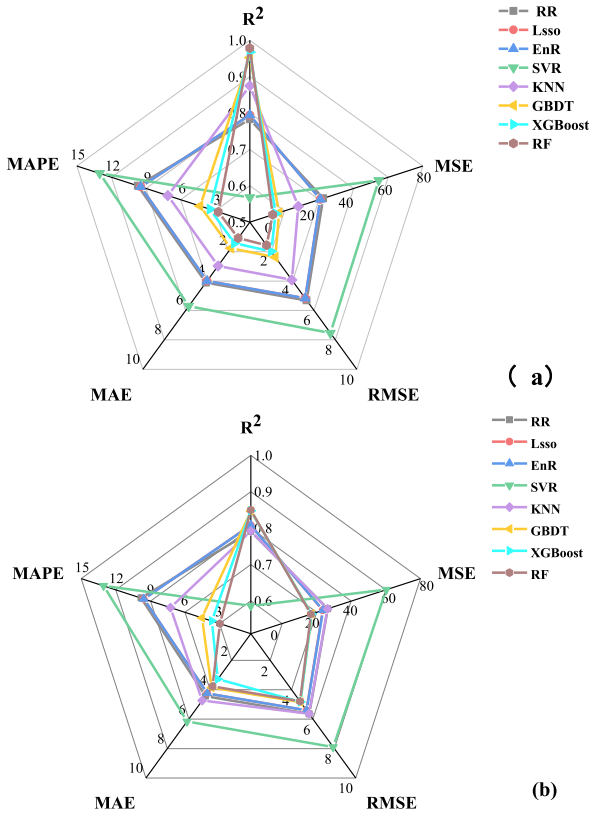


**Fig. 11.** Radar chart of the performance evaluation.
(a) Training data, (b) Testing data.

**Table 6**
Hyperparameter tuning scopes for model.

| Model | Hyper parameters | Description | Scope |
|---|---|---|---|
| GS-KNN | n-neighbors | Take the number of neighboring points k | [3,11,1] |
| GS-GBDT | n | Maximum number of iterations of the weak learner | [150,250,350] |
| | max-depth | Maximum depth of decision tree | [1,2,3] |
| | min-samples-split | Minimum number of samples required for internal node repartitioning | [5,6,7] |
| GS-XGBoost | n | Maximum number of iterations of the weak learner | [100,200,300,] |
| | max-depth | Maximum depth of decision tree | [1,2,3] |
| GS-RF | n | Maximum number of iterations of the weak learner | [100,150,200] |
| | max-features | Maximum number of features to consider when constructing the optimal model of decision tree | [8,12,16] |
| | min-samples-split | Minimum number of samples required for internal node repartitioning | [2,4,6] |

with a range of 3-11 and a step size of 1. GBDT has three hyperparameters: n-estimators, max-depth, and min-samples-split. The number of weak learners (n) represents the size of the model, and the depth of the tree affects the complexity of the model. Setting a minimum sample size (min-samples-split) prevents severe overfitting.

After optimizing the results through grid search, the KNN model shows excellent accuracy and generalization ability when the hyperparameter n-neighbors was set to 3. The optimal hyperparameters for the GBDT model were n = 350, max-depth = 2, and min-sample-split = 7. For the XGBoost model, the optimal hyperparameters were n = 200 and max-depth = 2. Finally, for the RF model, the optimal parameters are n = 200 and max-features = 8.

### *4.3.2. Optimization results*

Fig. 12 shows the relative distribution of predicted and true values of the model optimized by grid search, while Table 7 displays the evaluation metrics values of the optimized models. The chart clearly shows that the optimized model has a relatively concentrated distribution on the test set, with a significantly reduced deviation from the training set, performing better than the results in Figs. 7 and 9. The fit coefficients of the KNN, GBDT, XGBoost, and RF models in the test sets have all shown a significant improvement with an increase of 17%, 15%, 16%, and 15% respectively. Although grid search yielded the highest percentage improvement in test set accuracy for the KNN model among the four models, its fit $R^2$ value of 0.9270 is smaller than those of the other three models, indicating

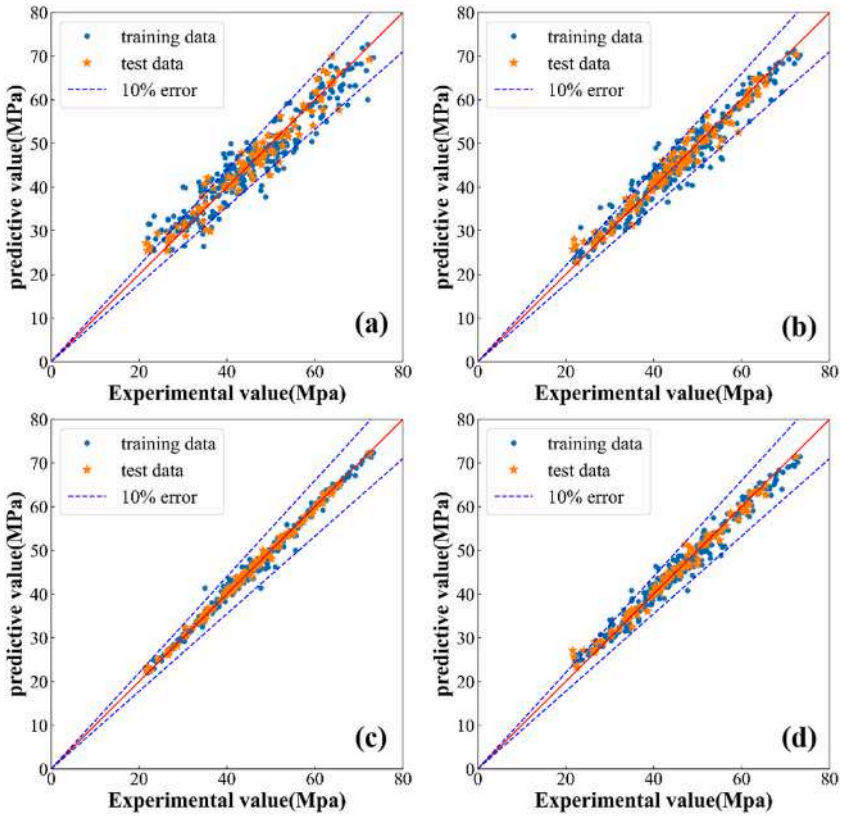


**Fig. 12.** Relationship between actual and predicted values of compressive strength for optimization models.

**Table 7**
Optimization modeling results in the numerical and relative systems.

| Model | Training data | | | | | Testing data | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | $R^2$ | MSE | RMSE | MAE | MAPE | $R^2$ | MSE | RMSE | MAE | MAPE |
| KNN | 0.9126 | 11.5702 | 3.4015 | 3.4015 | 6.1709 | 0.9270 | 10.9435 | 2.8576 | 2.1622 | 6.1708 |
| GBDT | 0.9855 | 1.9188 | 1.3852 | 0.9902 | 2.3132 | 0.9716 | 4.2569 | 2.1802 | 1.6632 | 1.6632 |
| XGBoost | 0.9927 | 0.9626 | 0.9811 | 0.6383 | 0.3018 | 0.9964 | 2.0587 | 0.7029 | 0.5255 | 1.4754 |
| RF | 0.9803 | 2.611 | 1.616 | 1.1575 | 2.8008 | 0.9862 | 3.1412 | 1.4142 | 1.0187 | 2.7915 |

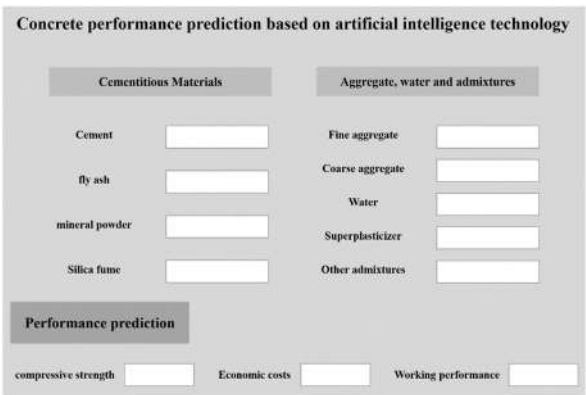


**Fig. 13.** Performance prediction platform for concrete materials.

that the KNN model still lags them. Among the other three models optimized by grid search, the GS-XGBoost model exhibits superior performance with a fitting coefficient of 0.9927 in the training set and outperforms its counterparts with smaller values of MSE (0.9626), RMSE (0.9811), MAE (0.6383) and MAPE (0.3018). Furthermore, it maintains its superiority in the test set with an $R^2$ value of 0.9964, while exhibiting lower values for MSE (2.0587), RMSE (0.7029), MAE (0.5255) and MAPE (1.4754). These data show that the grid search optimization algorithm can significantly improve the problem of poor generalization ability of basic machine algorithms and ensemble models, which is of great significance in the prediction of key performance of ordinary concrete.

### 4.4. Performance prediction platform

In this study, although a GS-XGBoost model was developed to predict the performance of plain concrete with high efficiency and accuracy, the lack of a simple graphic user interface (GUI) containing input and output information makes this technique more accessible to researchers [33]. To address this issue, a software for predicting the properties of ordinary concrete based on the GS-XGBoost model was developed in this section (shown in Fig. 13). This software effectively reduces the experimental and computational work required by researchers. As shown in Fig. 13, a graphical user interface (including two panels, input panel and output panel) can be used to predict the properties of ordinary concrete, such as compressive strength, economic cost, and workability. The database trained on the GS-XGBoost model in this study contains only the output variable compressive strength; training on economic cost and workability will be conducted in future studies. In addition, The database used in the study for development has certain limitations in terms of raw materials and performance, and in the future, the database will be continuously expanded to overcome the limitations and improve the prediction accuracy.

## 5. Conclusion and prospect

This study compares the application of different ML models (signal models and ensemble models) for the performance prediction of concrete materials and optimize these models by using grid search optimization algorithm. The results of the study showed that:

(1) The accuracy of single models varies greatly, with the most precise KNN model achieving an $R^2$ value exceeding 85%, while the least accurate SVR model falls below 60%. Furthermore, most models exhibit fit coefficients with $R^2$ values concentrated around 79%.
(2) The ensemble models exhibit high prediction accuracy, with $R^2$ values exceeding 95%. Among them, the XGBoost model achieves the highest accuracy ($R2 = 0.9984$). However, the fit coefficients of the three ensemble models have an $R^2$ below 0.85 and demonstrate low generalization performance.
(3) The increased complexity of the model increases its accuracy, as evidenced by higher fit coefficients for the ensemble model compare increases to those of the single model. This is mainly because the ensemble model combines the prediction results of multiple underlying models and takes advantage of the strengths of different underlying models to compensate for the deficiencies of single models. Therefore, the GBDT, XGBoost, RF models are most suitable for ordinary concrete strength prediction.
(4) After conducting grid search optimization, the KNN, GBDT, XGBoost, and RF models all exhibited significant enhancements in $R^2$ values on the test set with an increase of 17%, 15%, 16%, and 15% respectively. The GS-XGBoost model with the highest $R^2 > 0.95$ in both training and testing sets, exhibiting significantly improved generalization performance, demonstrates the effectiveness of grid search optimization.
(5) Finally, a software platform based on artificial intelligence technology has been developed for predicting the performance of concrete, including compressive strength, economic cost, and workability of input mix designs. This greatly reduces the amount of testing and calculations required by researchers and engineers.

This study innovatively explored an optimized integrated model algorithm that accurately predicts the compressive strength of ordinary concrete in actual engineering and established a prediction platform based on the optimal model. However, the concept of this work mainly focuses on the macro characteristics of concrete, of which input variables (mixture parameters) and output variable (only compressive strength) are relatively limited. In future work, we will consider adding automatic algorithms, such as AutoML. In addition, factors such as the composition and microstructure of concrete can be continuously considered in the model, further studying the feasibility of coordinated research on the composition, structure, and performance of concrete materials, and building robust models with strong extrapolation capabilities based on this.

## Author statement

**Kexin Liu:** Formal analysis, Data curation, Investigation, Methodology, Writing-original draft.
**Lingyan Zhang:** Formal analysis, Data curation, Methodology, Investigation, Writing - review & editing.
**Weiguang Wang**: Formal analysis, Data curation.
**Guozhi Zhang**: Conceptualization, Investigation.
**Liuliu Xu**: Formal analysis, Investigation.
**Dingqiang Fan**: Conceptualization, Validation, Supervision, Writing - review & editing.
**Rui Yu:** Conceptualization, Funding acquisition, Supervision, Writing - review & editing.

## Declaration of competing interest

There is no conflict of interest in this paper.

## Data availability

Data will be made available on request.

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## References

[1] N. Makul, Advanced smart concrete - a review of current progress, benefits and challenges, J. Clean. Prod. 274 (2020), 122899.
[2] J. Yang, R. Chen, Z. Zhang, Y. Zou, J. Zhou, J. Xia, Experimental study on the ultimate bearing capacity of damaged RC arches strengthened with ultra-high performance concrete, Eng. Struct. 279 (2023), 115611.
[3] K. Liu, T. Yin, D. Fan, J. Wang, R. Yu, Multiple effects of particle size distribution modulus (q) and maximum aggregate size (Dmax) on the characteristics of Ultra-High Performance concrete (UHPC): experiments and modeling, Cement Concr. Compos. 133 (2022), 104709.
[4] M. Sun, R. Yu, C. Jiang, D. Fan, Z. Shui, Quantitative effect of seawater on the hydration kinetics and microstructure development of Ultra High Performance Concrete (UHPC), Construct. Build. Mater. 340 (2022), 127733.
[5] K. Devi, P. Aggarwal, B. Saini, Admixtures used in self-compacting concrete: a review, Iranian Journal of Science and Technology, Transactions of Civil Engineering 44 (2) (2020) 377–403.
[6] M.R. Hossain, R. Sultana, M.M. Patwary, N. Khunga, P. Sharma, S.J. Shaker, Self-healing concrete for sustainable buildings. A review, Environ. Chem. Lett. 20 (2) (2022) 1265–1273, https://doi.org/10.1007/s10311-021-01375-9.
[7] Y. Zou, K. Zheng, Z. Zhou, Z. Zhang, J. Guo, J. Jiang, Experimental study on flexural behavior of hollow steel-UHPC composite bridge deck, Eng. Struct. 274 (2023), 115087.
[8] Z. Zhang, K. Pang, L. Xu, Y. Zou, J. Yang, C. Wang, The bond properties between UHPC and stone under different interface treatment methods, Construct. Build. Mater. 365 (2023), 130092.
[9] X.Y. Zhang, M.X. Fan, Y.X. Zhou, D.D. Ji, J.H. Li, R. Yu, Development of a sustainable alkali activated ultra-high performance concrete (A-UHPC) incorporating recycled concrete fines, J. Build. Eng. 67 (2023), 105986.
[10] Y. Leng, Y. Rui, S. Zhonghe, F. Dingqiang, W. Jinnan, Y. Yonghuan, L. Qiqing, H. Xiang, Development of an environmental Ultra-High Performance Concrete (UHPC) incorporating carbonated recycled coarse aggregate, Construct. Build. Mater. 362 (2023), 129657.
[11] V. Vishwakarma, D. Ramachandran, Green Concrete mix using solid waste and nanoparticles as alternatives – a review, Construct. Build. Mater. 162 (2018) 96–103.
[12] M. Sun, R. Yu, C. Jiang, D. Fan, Z. Shui, Quantitative effect of seawater on the hydration kinetics and microstructure development of Ultra High Performance Concrete (UHPC), Construct. Build. Mater. 340 (2022), 127733.
[13] X. Wang, R. Yu, Z. Shui, Q. Song, Z. Liu, M. Bao, Z. Liu, S. Wu, Optimized treatment of recycled construction and demolition waste in developing sustainable ultra-high performance concrete, J. Clean. Prod. 221 (2019) 805–816.
[14] Z. Wang, Z. Shui, T. Sun, X. Li, M. Zhang, Recycling utilization of phosphogypsum in eco excess-sulphate cement: synergistic effects of metakaolin and slag additives on hydration, strength and microstructure, J. Clean. Prod. 358 (2022), 131901.
[15] D.Q. Fan, Z.H. Shui, R. Yu, X.P. Wang, X.H. Li, Preparation of eco-friendly ultra-high performance concrete by lead-zinc tailings, Bull. Chin. Ceram. Soc. 37 (2018) 2231–2236.
[16] N. Narayanan, K. Ramamurthy, Structure and properties of aerated concrete: a review, Cement Concr. Compos. 22 (5) (2000) 321–329.
[17] Y. Zou, K. Yu, J. Heng, Z. Zhang, H. Peng, C. Wu, X. Wang, Feasibility study of new GFRP grid web-Concrete composite beam, Compos. Struct. 305 (2023), 116527.
[18] Z. Wen, R. Zhou, H. Su, MR and stacked GRUs neural network combined model and its application for deformation prediction of concrete dam, Expert Syst. Appl. 201 (2022), 117272.
[19] G. Boulton-Lewis, T. Cooper, B. Atweh, H. Pillay, L. Wilss, S. Mutch, Processing load and the use of concrete representations and strategies for solving linear equations, J. Math. Behav. 16 (4) (1997) 379–397.
[20] M.A. DeRousseau, J.R. Kasprzyk, W.V. Srubar, Computational design optimization of concrete mixtures: a review, Cement Concr. Res. 109 (2018) 42–53.
[21] F. Dingqiang, R. Yu, L. Kangning, T. Junhui, S. Zhonghe, W. Chunfeng, W. Shuo, G. Zhenfeng, H. Zhengdong, S. Qiqi, Optimized design of steel fibres reinforced ultra-high performance concrete (UHPC) composites: towards to dense structure and efficient fibre application, Construct. Build. Mater. 273 (2021), 121698.
[22] I. Ferdosian, A. Camões, Eco-efficient ultra-high performance concrete development by means of response surface methodology, Cement Concr. Compos. 84 (2017) 146–156.
[23] M. Shafikhani, S.E. Chidiac, Quantification of concrete chloride diffusion coefficient – a critical review, Cement Concr. Compos. 99 (2019) 225–250.
[24] A. Habibi, A.M. Ramezanianpour, M. Mahdikhani, O. Bamshad, RSM-based evaluation of mechanical and durability properties of recycled aggregate concrete containing GGBFS and silica fume, Construct. Build. Mater. 270 (2021), 121431.
[25] Z.S.O.A. Hui Ma, Research on compressive strength of manufactured sand concrete based on response surface methodology (RSM), J. Appl. Sci. 7 (2022) 1–13.
[26] D. Fan, R. Yu, Z. Shui, K. Liu, Y. Feng, S. Wang, K. Li, J. Tan, Y. He, A new development of eco-friendly Ultra-High performance concrete (UHPC): towards efficient steel slag application and multi-objective optimization, Construct. Build. Mater. 306 (2021), 124913.
[27] V. Sergis, C.M. Ouellet-Plamondon, D-optimal design of experiments applied to 3D high-performance concrete printing mix design, Mater. Des. 218 (2022), 110681.
[28] B. Yu, Q. Ma, H. Huang, Z. Chen, Probabilistic prediction model for chloride diffusion coefficient of concrete in terms of material parameters, Construct. Build. Mater. 215 (2019) 941–957.
[29] R. Homayoonmehr, A. Rahai, R.A. Akbar, Predicting the chloride diffusion coefficient and surface electrical resistivity of concrete using statistical regression-based models and its application in chloride-induced corrosion service life prediction of RC structures, Construct. Build. Mater. 357 (2022).
[30] J.N.K.K. Steven, L. Brunton, Data-Driven aerospace engineering: reframing the industry with machine learning, AIAA J. 59 (2021) 2820–2847.
[31] J.T. McCoy, L. Auret, Machine learning applications in minerals processing: a review, Miner. Eng. 132 (2019) 95–109.
[32] H. Sun, H.V. Burton, H. Huang, Machine learning applications for building structural design and performance assessment: state-of-the-art review, J. Build. Eng. 33 (2021), 101816.

[33] D. Fan, R. Yu, S. Fu, L. Yue, C. Wu, Z. Shui, K. Liu, Q. Song, M. Sun, C. Jiang, Precise design and characteristics prediction of Ultra-High Performance Concrete (UHPC) based on artificial intelligence techniques, Cement Concr. Compos. 122 (2021), 104171.
[34] H. Jia, G. Qiao, P. Han, Machine learning algorithms in the environmental corrosion evaluation of reinforced concrete structures - a review, Cement Concr. Compos. 133 (2022), 104725.
[35] D. Fan, J. Zhu, M. Fan, J. Lu, S.H. Chu, E. Dong, R. Yu, Intelligent design and manufacturing of ultra-high performance concrete (UHPC)–A review, Construct. Build. Mater. 385 (2023), 131495.
[36] B.A. Salami, M. Iqbal, A. Abdulraheem, F.E. Jalal, W. Alimi, A. Jamal, T. Tafsirojjaman, Y. Liu, A. Bardhan, Estimating compressive strength of lightweight foamed concrete using neural, genetic and ensemble machine learning approaches, Cement Concr. Compos. 133 (2022), 104721.
[37] H. Nguyen, T. Vu, T.P. Vo, H. Thai, Efficient machine learning models for prediction of concrete strengths, Construct. Build. Mater. 266 (2021), 120950.
[38] A. Hammoudi, K. Moussaceb, C. Belebchouche, F. Dahmoune, Comparison of artificial neural network (ANN) and response surface methodology (RSM) prediction in compressive strength of recycled concrete aggregates, Construct. Build. Mater. 209 (2019) 425–436.
[39] W. Zhao, S. Feng, J. Liu, B. Sun, An explainable intelligent algorithm for the multiple performance prediction of cement-based grouting materials, Construct. Build. Mater. 366 (2023), 130146.
[40] Y. Wu, Y. Zhou, Hybrid machine learning model and Shapley additive explanations for compressive strength of sustainable concrete, Construct. Build. Mater. 330 (2022), 127298.
[41] A. Abdulalim Alabdullah, M. Iqbal, M. Zahid, K. Khan, M. Nasir Amin, F.E. Jalal, Prediction of rapid chloride penetration resistance of metakaolin based high strength concrete using light GBM and XGBoost models by incorporating SHAP analysis, Construct. Build. Mater. 345 (2022), 128296.
[42] G.B. T, Standard for Test Methods of Concrete Physical and Mechanical Properties, China Architecture & Building Press Beijing, China, 2019.
[43] T. GB, 50080-2016, Standard for Test Method of Performance on Ordinary Fresh Concrete, China Architecture and Building Press, Beijing, 2016.
[44] Y. Li, J. Shen, H. Lin, H. Li, J. Lv, S. Feng, J. Ci, The data-driven research on bond strength between fly ash-based geopolymer concrete and reinforcing bars, Construct. Build. Mater. 357 (2022), 129384.
[45] S. Mahjoubi, W. Meng, Y. Bao, Auto-tune learning framework for prediction of flowability, mechanical properties, and porosity of ultra-high-performance concrete (UHPC), Appl. Soft Comput. 115 (2022), 108182.
[46] M.R. Kaloop, D. Kumar, P. Samui, J.W. Hu, D. Kim, Compressive strength prediction of high-performance concrete using gradient tree boosting machine, Construct. Build. Mater. 264 (2020), 120198.
[47] F. Aslam, F. Furqan, N.A.K.K. Muhammad, A. Arslan, F.J. Muhammad, A. Rayed, H.A. And, Applications of gene expression programming for estimating compressive strength of high-strength concrete, Adv. Civ. Eng. 2020 (2020) 1–23.
[48] T. Chai, R.R. Draxler, Root mean square error (RMSE) or mean absolute error (MAE)?–Arguments against avoiding RMSE in the literature, Geosci. Model Dev. (GMD) 7 (2014) 1247–1250.
[49] S.F.A. Shah, B. Chen, M. Zahid, M.R. Ahmad, Compressive strength prediction of one-part alkali activated material enabled by interpretable machine learning, Construct. Build. Mater. 360 (2022), 129534.
[50] P.R. Prem, A.M. Vermaa, Applied linear and nonlinear statistical models for evaluating strength of Geopolymer concrete, Comput. Concr. 24 (2019) 7–17.
[51] M. Kang, D. Yoo, R. Gupta, Machine learning-based prediction for compressive and flexural strengths of steel fiber-reinforced concrete, Construct. Build. Mater. 266 (2021), 121117.
[52] K.J.T. Elevado, G.G.A.R. Joenel, Compressive strength modelling of concrete mixed with fly ash and waste ceramics using K-nearest neighbor algorithm, Geotec. Const. Mat. & Env. 15 (2018) 169–174.
[53] Q. Li, Z. Song, High-performance concrete strength prediction based on ensemble learning, Construct. Build. Mater. 324 (2022), 126694.
[54] J. Feng, H. Zhang, K. Gao, Y. Liao, W. Gao, G. Wu, Efficient creep prediction of recycled aggregate concrete via machine learning algorithms, Construct. Build. Mater. 360 (2022), 129497.
[55] S.L.Y.C. Zhengyu Fei, Ensemble machine-learning-based prediction models for the compressive strength of recycled powder mortar, Materials 16 (2023) 583.
[56] H.A. Alkadhim, N.A.W.A. Muhammad, K. Kaffayatullah, S. Nazar, I.F.A.M. Muhammad, Evaluating the strength and impact of raw ingredients of cement mortar incorporating waste glass powder using machine learning and SHapley additive, Materials 15 (2022) 7344.
[57] B.P. Koya, S. Aneja, R. Gupta, C. Valeo, Comparative analysis of different machine learning algorithms to predict mechanical properties of concrete, Mech. Adv. Mater. Struct. 29 (25) (2022).
[58] M. Li, Q. Dai, P. Su, Z. You, Y. Ma, Surface layer modulus prediction of asphalt pavement based on LTPP database and machine learning for Mechanical-Empirical rehabilitation design applications, Construct. Build. Mater. 344 (2022), 128303.
[59] A. Géron, Hands-on Machine Learning with Scikit-Learn, Keras, and TensorFlow, O'Reilly Media, 2022.
[60] B.P.B.T. Mr Osborne, New approach to variable selection in least squares problems, Journal of Numeical Analysis (2000) 1996, 2023 (2023-4-20).
[61] C.J. Ranstam J, LASSO Regression, 2018, 1348.
[62] S.L.G.Z. Weizhang Liang, Predicting Hard Rock Pillar Stability Using GBDT, XGBoost, and LightGBM Algorithms, 2020.
[63] Z. Yang, W. Gao, L. Chen, C. Yuan, Q. Chen, Q. Kong, A novel electromechanical impedance-based method for non-destructive evaluation of concrete fiber content, Construct. Build. Mater. 351 (2022), 128972.
[64] Z. Fan, Evaluation of feature selection methods using bagging and boosting ensemble techniques on high throughput biological data | proceedings of the 2020 10th international conference on biomedical engineering and technology, in: International Conference On Biomedical Engineering and Technology 2020, 2020, pp. 170–175.
[65] J. Duan, P.G. Asteris, H. Nguyen, X. Bui, H. Moayedi, A novel artificial intelligence technique to predict compressive strength of recycled aggregate concrete using ICA-XGBoost model, Eng. Comput. 37 (4) (2021) 3329–3346.
[66] W. Huo, Z. Zhu, H. Sun, B. Ma, L. Yang, Development of machine learning models for the prediction of the compressive strength of calcium-based geopolymers, J. Clean. Prod. 380 (2022), 135159.
[67] A. Parmar, R. Katariya, V. Patel, A review on random forest: an ensemble classifier, International Conference on Intelligent Data Communication Technologies and Internet of Things (ICICI) 2018 (2019) 758–763.
[68] D. Liu, Z. Fan, Q. Fu, M. Li, M.A. Faiz, S. Ali, T. Li, L. Zhang, M.I. Khan, Random forest regression evaluation model of regional flood disaster resilience based on the whale optimization algorithm, J. Clean. Prod. 250 (2020), 119468.
[69] F. Tadayoshi, Estimation of prediction error by using K-fold cross-validation, Stat. Comput. 21 (2) (2011).
[70] D. Juan Rodriguez, Aritz Perez, A. Jose Lozano, Sensitivity analysis of k-Fold Cross validation in prediction error estimation, IEEE Trans. Pattern Anal. Mach. Intell. 32 (3) (2010).
[71] B. Yoshua, G. Yves, No unbiased estimator of the variance of K-Fold Cross-Validation, J. Mach. Learn. Res. 5 (2004).
[72] T. Wong, P. Yeh, Reliable accuracy estimates from *k*-Fold Cross validation, Ieee Xplore 1 (2019) 20, 2023 (2023-4-20).
[73] Y.M.J.O. Zaineb, M. Alhakeem, Prediction of ecofriendly concrete compressive strength using gradient boosting regression tree combined with GridSearchCV hyperparameter-optimization techniques, 2023 (2023-4-6), Materials (2022) 164. and superplasticizer (kg). The root mean square error (RMSE) and coefficient of determination (R2, with RMSE and R2.
[74] Z. Yang, W. Gao, L. Chen, C. Yuan, Q. Chen, Q. Kong, A novel electromechanical impedance-based method for non-destructive evaluation of concrete fiber content, Construct. Build. Mater. 351 (2022), 128972.
[75] M. Su, Q. Zhong, H. Peng, S. Li, Selected machine learning approaches for predicting the interfacial bond strength between FRPs and concrete, Construct. Build. Mater. 270 (2021), 121456.
[76] P.M.P.M. Schober, C.P.M. Boer, L.A.M.P. Schwarte, Correlation coefficients: appropriate use and interpretation : anesthesia & analgesia, Anesth. Analg. 2 (1) (2018) 1, 2023 (2023-4-20).
[77] F. Kuang, Z. Long, D. Kuang, X. Liu, R. Guo, Application of back propagation neural network to the modeling of slump and compressive strength of composite geopolymers, Comput. Mater. Sci. 206 (2022), 111241.

[78] M.A. Khalaf, C.B. Cheah, M. Ramli, N.M. Ahmed, A. Al-Shwaiter, Effect of nano zinc oxide and silica on mechanical, fluid transport and radiation attenuation properties of steel furnace slag heavyweight concrete, Construct. Build. Mater. 274 (2021), 121785.
[79] K. Liu, T. Yin, D. Fan, J. Wang, R. Yu, Multiple effects of particle size distribution modulus (q) and maximum aggregate size (Dmax) on the characteristics of Ultra-High Performance concrete (UHPC): experiments and modeling, Cement Concr. Compos. 133 (2022), 104709.
[80] K.J.T. Elevado, G.G.A.R. Joenel, Compressive strength modelling of concrete mixed with fly ash and waste ceramics using K-nearest neighbor algorithm, Int. J. GEOMATE 15 (2018) 169–174.
[81] K. Muthusamy, J. Mirza, N.A. Zamri, M.W. Hussin, A.P.P. Abdul Majeed, A. Kusbiantoro, A.M. Albshir Budiea, Properties of high strength palm oil clinker lightweight concrete containing palm oil fuel ash in tropical climate, Construct. Build. Mater. 199 (2019) 163–177.
[82] H. Huang, H. Lei, S.F.J. Rongjia, Bus single-trip time prediction based on ensemble learning, Comput. Intell. Neurosci. 1 (2) (2022).
[83] K. Yang, Z. Yu, C.L.P. Chen, W. Cao, Incremental weighted ensemble broad learning system for imbalanced data, IEEE Trans. Knowl. Data Eng. 34 (2022) 5809–5824.
[84] Y. Bian, H. Chen, When does diversity help generalization in classification ensembles? Ieee T. Cybern. 52 (2022) 9059–9075.